Journal of Cleaner Production 426 (2023) 139015

Contents lists available at ScienceDirect

Journal of Cleaner Production

journal homepage: www.elsevier.com/locate/jclepro

Analyzing the effect of normally distributed cooling channels on a photovoltaic thermal solar unit

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ARTICLE INFO

Handling Editor: Panos Seferlis

Keywords:
PVT
Electrical efficiency
Thermal efficiency
Dual power generation
GTO
RBFNN

ABSTRACT

Photovoltaic thermal (PVT) systems convert solar radiation into electrical and thermal energy while keeping the PV's temperature within a permissible range. This paper presents a numerical analysis of a novel normal cooling technique for the PV module and a new AI prediction model. It considers a cooling direction normal to the PV panel using rectangular cascade channels with two different orientations. This cascading of the flow inlet channels helps keep the inflow temperature as cold as possible along the PV panel. The numerical simulations are performed based on a flow rate range of 0.00024: 0.0012 LPM, insolation of 400: 1200 W/m^2, convection coefficients of 5.48 and 20.7 W/m^2.K, and ambient temperatures of 298 and 308 K. The proposed PVT system achieves a decrease in the cell's average temperature by 27K while improving the electrical, thermal, and overall efficiencies by 4.8%,59.5%, and 73.33%, respectively. The numerical database is used for building an ANN prediction model based on a radial base function neural network with Gorilla troops optimizer (RBFNN-GTO). The proposed RBFNN-GTO model is developed to predict the main performance parameters of the PVT which are thermal efficiency, electrical efficiency, overall efficiency, average temperature of the PV panel, and the model temperature un-uniformity. It achieves an overall correlation coefficient of 0.9997 concerning the numerical database and showed an average MSE of 5.4313E-4.

1. Introduction

PV systems are one of the widely used solar energy harvesting systems due to their ease of installation and maintenance. One of the problems found when using PV systems is the deterioration of its efficiency with the increase of its temperature. Bianchini et al. (2017) and (Kalogirou and Tripanagnostopoulos, 2006) emphasized that the electrical efficiency (η_{el}) of the PV panels decreases by (0.3–0.5%) when its temperature increases by 1 °C. Therefore, a branch of investigation has been carried out to cool the PV systems and enhance the heat transferred from them.

Several techniques were proposed and investigated for PV cooling and integrating the PV with thermal modules (PVT) for producing both thermal and electrical energy. These approaches increased the PV overall efficiency (η_{ov}) dramatically. In the PVT systems, the effect of the cooling is greater for harvesting thermal energy and has little enhancement effect on the electrical efficiency (η_{el}). With the PVT, the overall efficiency (η_{ov}) reached up to 80% (De Luca et al., 2023). The cooling of the PV modules was conducted by active and passive techniques. The active techniques consider forcing the flow to produce forced convective cooling. However, passive techniques consider natural convection. Although active techniques consume pumping power, which affects η_{ov}, it exerts better cooling for the PV module and enlarges the produced thermal power. De Luca et al. (2023) investigated the effect of a vacuum layer inside the encapsulation of the PVT module. They used two configurations for the module. The first module applied the vacuum layer beneath the glass directly. The second configuration applied the vacuum layer beneath the PV cells itself. They found that the vacuum has a noticeable effect of reducing the thermal losses by convection. The energy efficiency reached higher than 80% for both configurations. Bayrak et al. (2019) investigated experimentally the cooling effect of aluminum fins fixed on the backside of the PV module with ten different configurations as passive cooling. They found that the best

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https://doi.org/10.1016/j.jclepro.2023.139015
Received 17 June 2023; Received in revised form 11 September 2023; Accepted 24 September 2023
Available online 2 October 2023
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Abbreviations

ACO	Ant colony optimization
AI	Artificial intelligence
ANFIS	adaptive neuro-fuzzy interface system
ANN	Artificial neural network
ARE	absolute relative error ( )
ELM	Extreme learning machine
FFNN	feedforward neural network
FMCHS	fractal microchannel heat sink
GR	Generalized regression
GTO	Gorilla troops optimizer
HTF	heat transfer fluid
HCPV	High concentration PV
ML	machine learning
MAPE	mean absolute percentage error
MSE	Mean square error
PCM	Phase change material
PV	Photovoltaic
PVT	photovoltaic thermal
RBFNN	Radial bias function neural network
SVMs	Support vector machines
SPF	Seasonal performance factor

Nomenclature

A	the area of the PV module (m^2)
B_{max}	Maximum value of GTO variable
B_{min}	Minimum value of GTO variable
Cp	specific heat capacity (kJ/kg K)
D	diameter (mm)
F	function
g	gravitational acceleration (m/s^2)
$GL(i)$	Vector solutions of gorilla locations
I	total number of hidden neurons
K	thermal conductivity (W/m. K)
K_t	total number of gorillas
k	Turblence kinetic energy
L_{sb}	Silverback best location
N	Number of training patterns
P	Pressure (N/m^2)
$\dot{Q}$	volume flow rate [m^3/s]
t	the transmissivity ( )
T	Temperature (K)
u	fluid velocity (m/s)
$\overline{X}$	neural network input vector
X_r	randomly chosen gorilla
Y	neural network output
Z_h	RBFNN output

Greek symbols

Δ	Difference
η	Efficiency
γ	the temperature coefficient of the PV module
ρ	Density (kg/m^3)
τ	shear stress in the fluid (Pa)
μ	fluid dynamic viscosity (Pa.s)
α	the absorptivity ( )
φ	Radial bias function
δ	Centre of the RBF
σ	Width of the RBF
ω	Turbulence specific dissipation rate
$\overline{\omega}$	Weighted factors

Subscripts

c	cell
el	electrical
f	fluid
fr	friction
g	glass
h	horizontal
ref	reference
th	Thermal
ove	Overall
obj	Objective
s	solid

energy and exergy efficiency of the PV module were 11.55% and 10.91%, respectively. Elminshawy et al. (2019) studied the performance of an air-cooled PV module. They used underground cooling for the air to achieve lower air temperatures. This technique decreased the temperature of the PV module by 13 °C and enhanced its efficiency by 18.9%. Shen et al. (2021) studied numerically the performance of a PVT system that contained a special cooling channel designed based on the field synergy theory. A mathematical model was validated experimentally for the channels with different types of fins. The results of this research revealed that η_{el} increases as the cooling water temperature decreases, and the water mass flow rate increases under high solar radiation. Furthermore, the Shark dorsal shape achieved better cooling than the V-shape, due to the smaller angle between the velocity gradient and the temperature gradient fields. Hassan et al. (2023) examined numerically and experimentally different parameters to enhance the heat transfer of a PVT system using water as a heat transfer fluid (HTF). These parameters were tube diameters with different flow rates which in turn affected the Reynolds number and back plate thickness. They found that the heat transfer, the thermal efficiency (η_{th}), and η_{el} increase as the Reynolds number increases. Their system achieved an η_{ov} of 59.3%. Kalateh et al. (2022) studied experimentally the effect of twisted tapes inserted inside the riser tubes of the collector located back of the PVT system. Clockwise and counter-clockwise twisted tapes were used with water as HTF to cool the PVT. By using twisting tapes, the surface temperature of the system decreased under different values of solar intensity. Furthermore, the energy efficiency increased for the PVT system by 28.4%. In addition, higher exergy efficiency and lower entropy generation were achieved. Baloch et al. (2015) investigated numerically and experimentally the performance of a cooled PV system using heat exchangers with converging channels at different angles of converge. A comparison between the uncooled and cooled PV systems showed a significant reduction in the cell temperature. In addition, the generated power, and the efficiency for the cooled PV were improved by 35.5% and 36.1%, respectively, when compared with the uncooled one. The cost analysis revealed a cost reduction of 19.5% when using the cooled PV. Chiang et al. (2022) studied a loop thermosiphon experimentally to extract heat from a hybrid PVT system. The thermosiphon was located back to the PV system and used water as a cooling fluid. The results showed a reduction in the operating temperature by 17%, and an improvement in η_{el} by 7.3%. (Menon et al., 2022) evaluated experimentally the performance of an unglazed PVT system that was cooled by nanofluid and water running inside a setup configuration of the serpentine coil and thermal absorber tube. The panel's temperature decreased by 23.7 °C and 15 °C using nanofluid and water, respectively. Moreover, the values of η_{th} and η_{el} were increased significantly. The results revealed that η_{ov} of the nanofluid cooled is higher than that of water cooled by 21%. Shao et al. (2022) investigated experimentally a heat exchanger PVT ventilated roof for heat pump under different

Table 1
Review of studies conducted on predicting and modeling the performance of the PVT systems using ANN.

Reference	Input parameters	ANN types	Training algorithm	Output parameters	Performance function	Error value
Juwel Chandra Mojumder et al. (Mojumder et al., 2017)	Solar irradiance and ambient temperature	Extreme Learning Machine (ELM)	Gradient-based learning	Thermal and electrical efficiency	RMSE	0.001937
N. Gunasekar et al. (Gunasekar et al., 2015)	Solar intensity, ambient temperature, relative humidity, and wind velocity	FFNN	Back propagation	Evaporator heat gain, Solar energy input ratio, Panel efficiency (%), Panel temperature	RMSE	0.014%
Ahmed F. Mashaly et al. (Mashaly et al., 2015)	Air temperature, relative humidity, wind speed, solar radiation, ultraviolet index, temperature of feed water, temperature of brine, total dissolved solids of feed water, total dissolved solids of brine, and the number of days	FFNN	Back propagation	Distillate water productivity, operational recovery ratio and thermal efficiency	RMSR	0.0387 2.59 3.41
Kalani et al. (Kalani et al., 2017)	Ambient temperature, incident radiation, and fluid inlet temperature	FFNN RBFANN	Particle Swarm Optimization (PSO)	Fluid outlet temperature, electrical efficiency	RMSE MAPE	0.3748 0.7053 0.3235 0.6267
Ali H.A. Al-Waeli et al. (Al-Waeli et al., 2019)	Solar irradiation and ambient temperature,	FFNN	Back propagation	Current, Voltage, electrical efficiency, and thermal efficiency	MSE	0.0282
Neha Dimri et al. (Dimri et al., 2019)	Global solar radiation, and ambient air temperature	FFNN	Trial and error	Average fluid temperature, overall electrical efficiency, thermal efficiency, and overall exergy efficiency	root mean square percent deviation	1.86% 0.26% 0.85% 0.27%
Shahram Delfani et al. (Delfani et al., 2019)	GO nanofluid concentration flowrate, collector depth, collector length, and reduced temperature difference	FFNN	ML	Collector efficiency and Nusselt number	MAPE	2.208%
Ali H.A. Al-Waeli et al. (Al-Waeli et al., 2020)	Solar irradiance, and temperature ambient	MLP and Self-organizing Feature Map network	Momentum	Power (P)	MSE, RMSE	0.0223, 0.149
Chaibi et al. (Chaibi et al., 2021b)	Solar irradiance, module temperature, humidity, and wind	FFNN	Back propagation	Electrical efficiency, and thermal efficiency	MAPE	0.0078% 3.3607%
Yan Cao et al. (Cao et al., 2022)	Radiation intensity and nanofluid flow rate	ANFIS, Cascade feedforward, FFNN, Generalized regression, and RBFNN	Hybrid learning algorithm, ML algorithm	Electrical Efficiency (%)	RMSE	1.60 1.48 1.63 1.59 2.24
Orhan Büyükalaca et al. (Büyükalaca et al., 2023)	Flow rate, solar radiation, volumetric ratio of hBN nanoparticles, nano-particle diameter, and convective heat transfer coefficient.	FFNN	Backpropagation	Cell temperature, fluid outlet temperature, Electrical efficiency, Thermal efficiency, and exergy efficiency	RMSE	0.08230 °C 0.07460 °C 0.35540% 0.02370% 0.06390%
Sanjeev Jakhar et al. (Jakhar et al.)	Nanofluid concentration, Reynolds number, and time	FFNN	ML algorithms	Thermal and electrical performance of PV/T system	RMSE	–

conditions. This system was proposed to supply heat, generate power, and be weatherproof. The results indicated that the proposed system maintains the thermal stability of the environment and acceptable energy saving. The daily reduction in the heat gain reached 54.3%. Ahmadinejad and Moosavi (2023) evaluated numerically different PVT modules with different working fluids from exergy and energy perspectives under different working conditions. Three modules were proposed: conventional PV module, PVT using a collector with a simple channel, and PVT using a collector with a baffled channel. The considered HTFs were water, nanofluid CuO/water, and nanofluid CNT/water. The results showed that, using nanofluids with baffled channels increases the electrical and thermal powers by 14% and 2.12%, respectively. Han et al. (2021) established a model for heat transfer to compare η_{th}, η_{el} and η_{ov} of photovoltaic - solar thermal systems (PV-ST) that operate separately, with PVT – solar thermal systems (PVT-ST) in series. Various parameters were considered in this study. The results revealed that, under the conditions of high solar radiation and high ambient temperatures, the efficiencies of the PVT-ST system are higher than those of the PV-ST. Chhugani et al. (2023) studied one of the important achievements of the PVT system usage with the heat pump system. They compared the performance of the heat pump-PVT system with the air-cooled heat pump. The heat pump-PVT system achieved a higher seasonal performance factor (SPF) by 14%, compared to the air-cooled heat pump. Kazemian et al. (2023) developed a PVT system with a collector enhancer by covering the solar thermal collector partially with PV panels. This modification enhanced the output power by 31.24% and decreased the payback period by 35.07%, compared to the traditional PV module. Hamzat et al. (2021) presented a literature review of the cooling technologies of PV systems. They argued that, using nanofluid cooling can improve the system performance dramatically. The results of this review showed that nanofluid coolant could improve the efficiency of the PV by 60%. Khodadadi and Sheikholeslami (2021) evaluated numerically several models of PVT systems integrated with nano-enhanced phase change material (PCM). The effect of fins number, type of coolant, and different solar intensities were investigated. It was found that using fins reduces the PV temperature and enhances its thermal performance. Furthermore, the system with 18 fins achieved η_{ov} of 79.54%, which enhanced the efficiency by 9.47%, compared to the

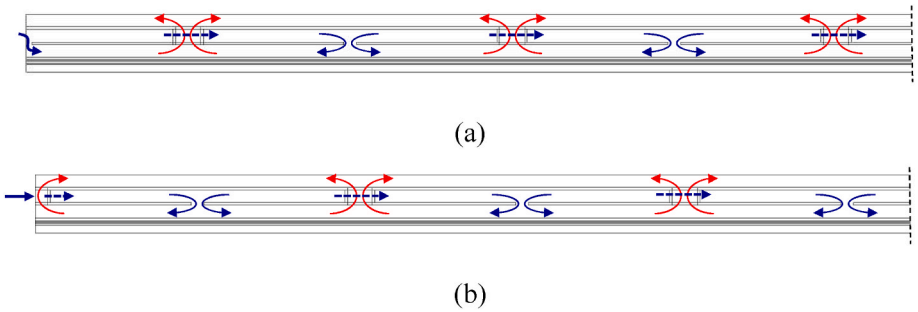


Fig. 1. Side view for the used configurations of multi-channel cooling for (a) Inlet-first geometry, and (b) Outlet-first geometry.

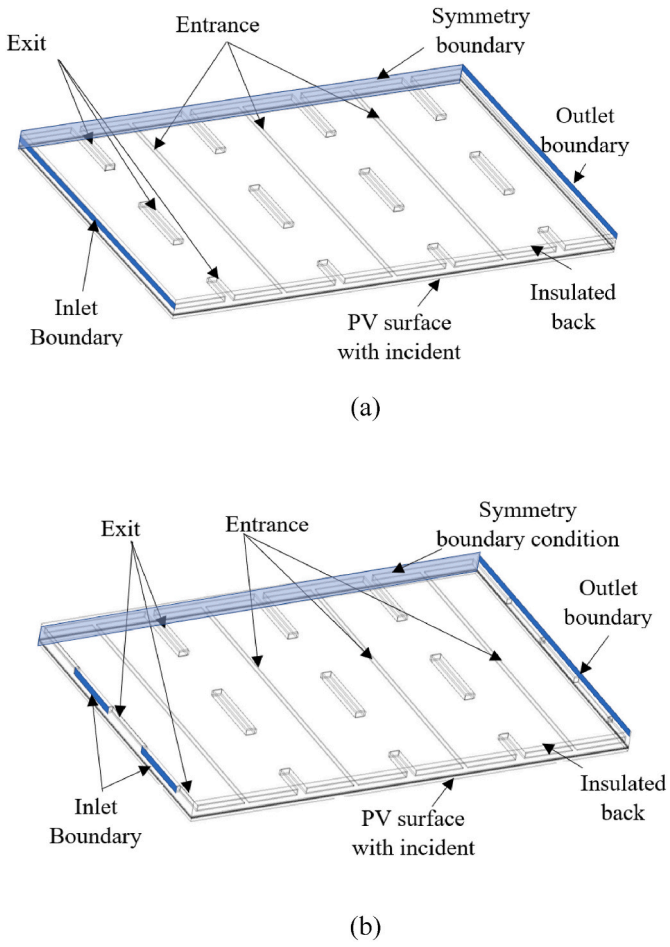


Fig. 2. The Thermal model geometry and boundary conditions for (a) inlet first model and (b) Outlet first model.

system without fins. Simón-Allué et al. (2022) addressed the performance of PVT with PCM, on the back side of the PV side and the thermal module. They found that, using PCM achieves an enhancement of 26% and 3% for η_{th} and η_{el}, respectively. Ul Abdin et al. (2022) analyzed a new PVT system with a storage tank located on the backside of the PV modules. The tank with electrical resistance is found to store thermal energy. A computer simulation technique was used to validate the suggested design usefulness and to study the influence of various parameters on the performance of the system. Song et al. (2023) established a simulated model of an air-heated PVT system with different arrangements of baffles. The best hole arrangement achieved an enhancement in the η_{th} and η_{el} by 4.3% and 12.9%, respectively. Choi and Choi (2022) discussed numerically the effect of different parameters on the efficiency and output energy of the PVT system using air as a coolant through a triangular collector. A mathematical model was developed. The results showed an enhancement in the energy outputs due to the selected design of the collector and the proper operating conditions. Moreover, both of η_{th} and η_{el} were enhanced by 36.97% and 2.59%, respectively. Peng et al. (2023) proposed a cooling method based on a double-layer tree-shaped fractal microchannel heat sink (FMCHS) for a high concentrator PV (HCPV) system. A model for flow and heat transfer was proposed. Each of η_{el}, the thermal performance, and the PV surface temperature were studied under different inlet velocities and flow modes. The designed double-layer FMCHS had higher cooling performance than the single-layer FMCHS and that the double-layer FMCHS achieved η_{el} up to 41.68%.

Recently, researchers have benefited from artificial intelligence (AI) technologies in modeling, predicting, controlling, and optimizing different processes in various fields (Khandakar et al., 2019). Artificial neural network (ANN) is one of the AI technologies that can find the relations between the input and output parameters of engineering problems. Moreover, it is used in modeling and predicting data. A hybrid technique based on machine learning (ML) methods was used to predict the output power from the PV module under different weather conditions (Maria and Yassine, 2020). It combined various methods of ML such as support vector machines (SVMs), discriminant analysis, classification ensembles, k-nearest neighbor, and classification trees algorithms. The recurrent neural network and ant lion optimization algorithm were applied to estimate the generated energy from the PV modules (Roy et al., 2019). It considered the variation in solar irradiance and the temperature as input parameters. Both η_{th} and η_{el} of the PVT system were estimated using the extreme LM method (Mojumder et al., 2017). A multi-layer perceptron was designed to determine η_{th} and η_{el} of the PVT systems based on design parameters such as Reynolds number, nanofluid concentration, and time (Jakhar et al.). The ANN was applied to estimate η_{th} and η_{el} of the PVT with air collector systems as well (Chaibi et al., 2021a). The PV module temperature and insolation were used as input parameters to the feedforward neural network (FFNN) for a PVT system cooled by water-based nanofluids to investigate the electrical performance (Cao et al., 2022). It used the trial-and-error method to obtain the best architecture of the ANN, least square SVM, and adaptive neuro-fuzzy interface system (ANFIS). Table 1 demonstrates a summary of studies using the ANN for modeling and predicting the performance of the PVT systems.

Many of the presented research works investigated the performance of the different geometrical configurations of the PVT. The flow direction and its contact with the PV module control its cooling efficiency. When the HTF remains in contact with the PV module, it is heated along its path. As it is heated, its cooling effect decreases due to the decrease in the temperature difference between the HTF and the PV back side.

In this paper, the contribution is found to be as follows.

- Presenting novel cascade channel cooling configuration to minimize the time through which the HTF remains in contact with the PV surface. So, the inlet fluid is divided into multi-channels to keep the

Table 2
PVT system materials and specifications.

Layer and material	Density [kg/m^3]	Thermal conductivity [W/m. K]	Cp [J/kg. k]	Thickness [mm]
Top Cover (Glass)	2450	2	500	3.2
Encapsulant (EVA)	950	0.311	2090	0.8
Solar cell (Silicon)	2329	148	700	0.5
Bottom cover (Tedlar)	1200	0.15	1250	0.1
Connector (Thermal paste)	2600	1.9	700	0.3
Metallic distributer (Stainless steel 316)	2700	160	900	1
Cooling space (water)	998	0.68	4200	20

Table 3
Guidelines for naming the different Conditions of the test cases.

Parameter Name	Parameter Range	Parameter Indicator in the Test Case Name
Channels' orientation	Inlet first	I
	Outlet first	O
Flow rate [LPM]	0.000245	F1
	0.00049	F2
	0.000735	F3
	0.00098	F4
	0.001225	F5
Solar radiation [W/m^2]	400	400
	600	600
	800	800
	1000	1000
	1200	1200
Convection coefficient [W/m^2.K]	5.84	C1
	20.7	C2
Ambient temperature [K]	298	T1
	303	T2

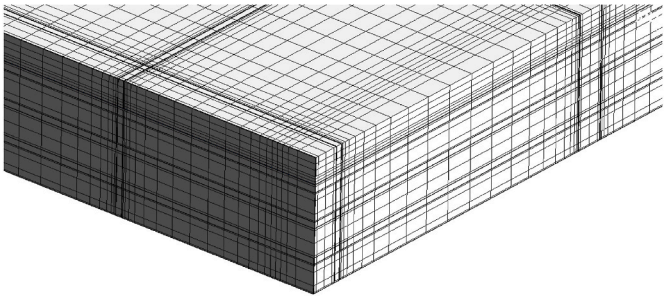


Fig. 3. The used mesh structure.

high-temperature difference between the HTF and the PV surface as high as possible.

- Two different configurations are considered. The difference between the two configurations is starting the flow over the thermal module with inlets or outlets channel passes.
- The proposed model is constructed of multi-rectangular channels as paths for the HTF with three vertical levels of HTF. The first level is responsible for entering the HTF, the second level, is in contact with the PV back side, and is the cooling level. The third level collects the hot HTF above the entrance level.
- The parameters considered are the HTF flow rate, the solar radiation intensity, the ambient temperature, and the convection coefficient. A CFD model was used for solving case studies with various parameters.
- A proposed RBFNN-GTO model presented for predicting η_{th}, $\eta_{el,}$ and η_{ov}, the average temperature of the PV panel, and the temperature un-uniformity factor of the PVT.
- Implementing the proposed GTO method to attain the optimum structure and parameters of the RBFNN.
- The proposed RBFNN-GTO model used solar irradiance, ambient temperature, convection coefficient, and mass flow rate of the cooling fluid as input parameters to predict the PVT performance.

2. Methodology

The proposed cooling system uses multi-inlets and outlets through cascade rectangular channels. The multi-inlets and outlets are proposed to keep the inlet flow temperature for each section as low as possible. This helps in improving the cooling and hence decreasing the PV panel's temperature. The proposed model side view for the two used geometries is shown in Fig. 1. The first geometry considers the inlet at the start of the PV module and is called Inlet-First geometry. The second geometry considers the heated water outlet at the start of the PV module and is called Outlet-First geometry. Both inlets and outlets in these geometries are considered concerning the PV module. The inlet slots are extended along the PV module width while the outlet slots take only a part of the width to permit the flow of the rest of HTF to pass to the next inlet slots. The 3D geometry is shown in more detail in Fig. 2.

2.1. Numerical investigation

In the present work, a new geometry for the thermal module applied in the PVT systems is tested. A numerical simulation is conducted to study the system performance with different parameters regarding its η_{th} and η_{el}. The PV panel considered in the present study is a 50 W poly-crystalline module EGE-50P-36. The dimension of the PV module is 520x680 × 30 mm. The PV module is considered as inclined by 45° on the horizontal. On the back side of the PV panel, a thermal module is fixed by a thermal paste. Ansys Fluent is used for CFD and heat transfer modeling, while Ansys ICEM software is used for building the geometry and mesh. SIMPLE Algorithm for velocity-pressure coupling is used. The standard pressure discretization scheme is considered, while the second-order upwind scheme is applied for the momentum and energy discretization. The first-order upwind scheme is considered for the turbulence equations discretization. An absolute convergence criterion of 10E-4 is considered for solving the continuity, momentum, and turbulence equations, while a value of 10E-6 is considered for the energy equation convergence. The layers of the PV module under study and its material properties are indicated in Table 2 (Nahar et al., 2017).

The rectangular passages are fixed inside the thermal module to guarantee multiple entrances and outlets for the cooling flow in contact with the PV panel. The cooling channels are inlet and outlet channels. The inlets are the continuous channels while the outlets are the discontinuous channels. Two cases are performed and tested in this study. In the first case, the inlet channels are set first, while in the second case, the outlet channels are set first. Both cases are illustrated in Fig. 2.

Table 3 indicates the conditions of the test cases considered in the present study according to the study parameters of the flow rate, the inlet and outlet locations, the convection coefficient, and the ambient temperature. These parameters are identified to keep the design cell temperature below 304K. The other study parameters are considered constant in the present work. The channels responsible for flow entrance to the PV back have a width of 5 mm. The exit channels have double this width (10 mm) as the length is half the panel's width only to permit the flow of the uncooled liquid to the other entrance channels on the rear side and keep the same area for the inlets and outlets. The name of each test followed the guidelines illustrated in Table 3. As a declaration, the name of the test "I–F1–400C1T1" can be considered as a test case with "Inlet first" channels orientation, the flowrate of 0.000245LPM, the ambient temperature of 303K, insolation of 400 W/m^2, convection

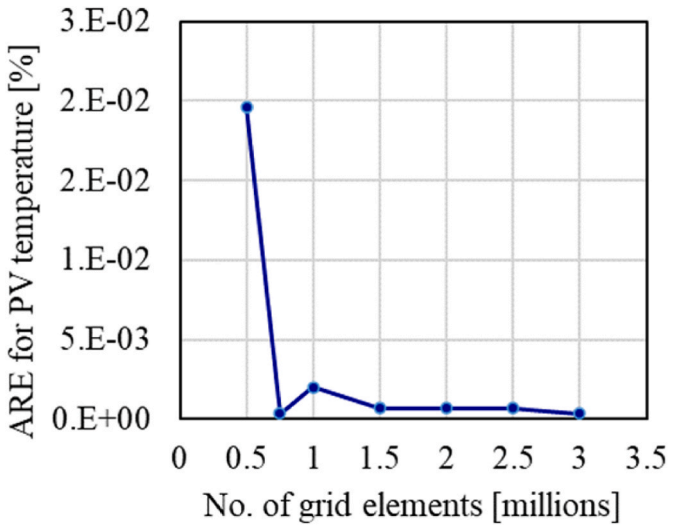


Fig. 4. Grid sensitivity ARE for the average top temperature of the PV module.

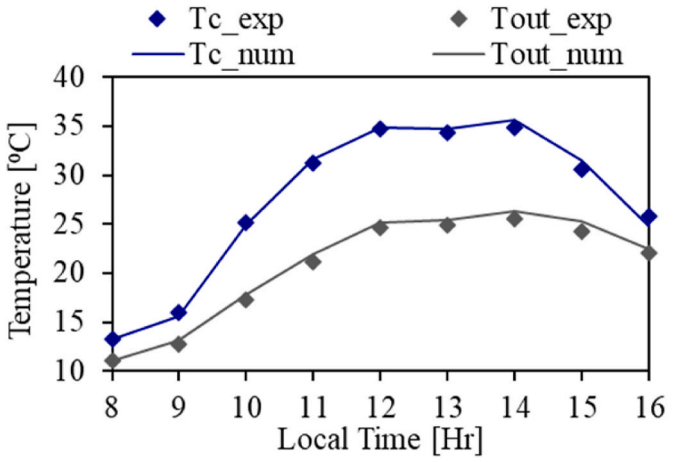


Fig. 5. Validation of the simulation with experimental data from Tiwari et al., 2006) (Tiwari et al., 2006).

coefficient of 5.84 W/m^2. K.

Ansys Fluent software is used in the present numerical study. It is based on the finite volume method. The present test case simulation is performed in steady-state condition.

2.1.1. Grid sensitivity test

A grid of 1.3 M structured hexahedral cells is created by Ansys ICEM for this study. The quality of the mesh used is analyzed and it ranges from 0.9 to 1, with skewness of 0.8: 1, and distortion of 0.95: 1. Small cell depth is constructed beside walls to account for turbulence beside walls considering the wall function requirements. A sample of the mesh used in this paper is shown in Fig. 3.

A mesh sensitivity analysis is performed for mesh sizes from 0.25 to 3 million elements. The flow rate is 0.001225 LPM under irradiance of 1200 W/m^2. The mesh-dependent solution is considered negligible after reaching a value of 10E-4 of the absolute relative error (ARE). The ARE considers the average PV temperature for the current and the last mesh relative to the current mesh. As can be observed from Fig. 4., after 1.5 million elements, the Absolute relative error (ARE) maintains around 6.6E-4%. So, this size is considered in all the simulations.

2.1.2. Model validation

The used model is compared with experimental results from Tiwari et al., 2006) (Tiwari et al., 2006). The PVT model used for validation has an area of 0.6 m^2 with a thermal model behind it in which cooling air flows. The model validation for the PV temperature and the outlet flow temperature is shown in Fig. 5. An acceptable agreement between the simulation and the experimental results is observed. The maximum ARE between the numerical and the experimental results is found to be 4.1%.

2.1.3. Governing equations

The equations solved in the present study are expressed as follows (Ansys Inc, 2017):

The continuity equation:

$$\frac{\partial(\rho\, u_i)}{\partial x_i} = 0 \tag{1}$$

The momentum equation:

$$\frac{\partial(\rho u_i u_j)}{\partial x_i} = -\frac{\partial P}{\partial x_i} + \frac{\partial}{\partial x_i}\left(\frac{\partial(\rho\, u_i)}{\partial x_i} - \rho\overline{\acute{u}_i \acute{u}_j}\right) + \rho\,\overrightarrow{g} \tag{2}$$

Energy equation in solid and liquid regions, respectively:

$$\frac{\partial}{\partial x_i}\left(k_s \frac{\partial T}{\partial x_i}\right) = 0 \tag{3}$$

$$\frac{\partial}{\partial x_i}(\rho u_i T) = \frac{\partial}{\partial x_i}\left(k_f \frac{\partial T}{\partial x_i}\right) \tag{4}$$

Although the flow speed is very low and the calculated Reynolds number is in the laminar flow domain, the geometry of the flow contains sharp edges that produce turbulence in the flow. Hence it is necessary to use turbulence modeling in this study. Turbulence modeling equations are used in the fluid domain for solving turbulence using the SST-k-ω model. The SST-k-ω model is based on the Baseline (BSL) k- ω model which applies the converted dissipation rate (ε) in the form of specific dissipation rate (ω) in the regions far from the wall. Near the wall, it applies the standard k- ω model with some modifications in the constants. This model solves the turbulence in the flow considering one equation for each of turbulence kinetic energy, k, and ω. The governing equations for the RNG-k- ω model can be expressed as follows.

$$\frac{\partial}{\partial x_i}(\rho k u_i) = \frac{\partial}{\partial x_i}\left(\Gamma_\kappa \frac{\partial k}{\partial x_i}\right) + G_k - Y_k + S_k \tag{5}$$

$$\frac{\partial}{\partial x_i}(\rho \omega u_i) = \frac{\partial}{\partial x_i}\left(\Gamma_\omega \frac{\partial \omega}{\partial x_i}\right) + G_\omega - Y_\omega + D_\omega + S_\omega \tag{6}$$

where Y_ω and Y_κ represent the turbulences produced by diffusion for ω and κ equations respectively. G_ω and G_κ are generations of turbulent kinetic energy due to mean velocity gradients. Γ_ω and Γ_κ represent the effective diffusivities, and S_ω and S_κ are the user-defined source terms, kg/(m s^3). The SST-k-ω is based on the BSL model with enhancements on the turbulent viscosity term.

2.2. Performance evaluation

2.2.1. Electrical model

The conversion of electromagnetic waves of the sun to electrical power takes place through the PV system. This conversion process is evaluated through the electrical conversion efficiency. The useful solar power entering the PV system, $P_{in,}$ [W] is defined as follows:

$$P_{in} = G * A \tag{7}$$

where G is the global inclined solar radiation power in [W/m^2], while A is the area of the PV module. The produced electrical power, P_{el} [W/m^2], is defined as follows:

$$P_{el} = \eta_{el} * P_{in} \tag{8}$$

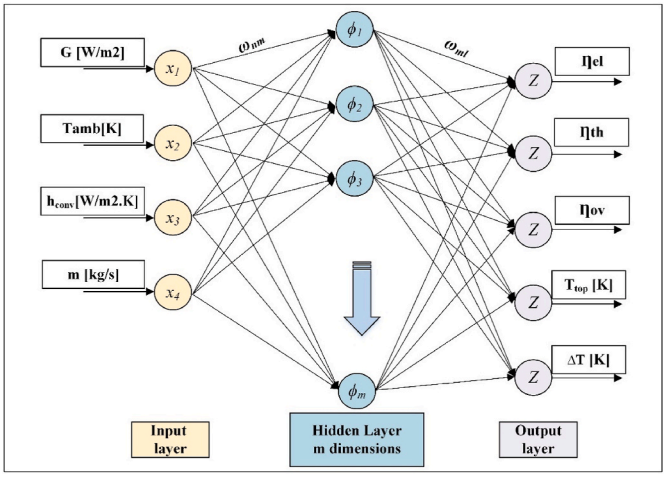


Fig. 6. Proposed RBFNN architecture.

where η_{el} is the electrical efficiency of the PV panel, which is defined as follows:

$$\eta_{el} = \eta_{ref}\left(1 - \gamma\left(T_c - T_{ref}\right)\right) \tag{9}$$

where η_{ref} is the reference efficiency of the PV module at the standard test conditions ($\eta_{ref} = 0.14$), γ is the temperature coefficient of the PV module ($\gamma = 0.0041/°C$), T_c is the cell temperature and T_{ref} is the reference temperature of the module ($T_{ref} = 25\ °C$).

2.2.2. Thermal model

In the thermal systems, the collected thermal power from the PVT system (P_{th}) is considered according to the thermal energy collected by the HTF. This value is calculated as follows:

$$P_{th} = \dot{m}\ C_p\ (T_{out} - T_{in}) \tag{10}$$

where $\dot{m}$ is the cooling fluid mass flow rate [kg/s], C_p is its specific heat in [J/kg.K], and $(T_{out} - T_{in})$ is the outlet and inlet fluid temperature difference in [K]. The thermal efficiency (η_{th}) is expressed as follows:

$$\eta_{th} = \frac{P_{th}}{P_{in}} \tag{11}$$

The variation of the flow rate affects the power consumed in the pumping of the HTF. This power is known as the friction power (P_{fr}) expressed as follows (Bahaidarah, 2016):

$$P_{fr} = \dot{Q} * \Delta p \tag{12}$$

where $\dot{Q}$ is the volume flow rate of the HTF, and Δp is the pressure drop encountered in the HTF between the inlet and the outlet.

The η_{ov} can be calculated from the summation of the electric and thermal power output. However, some power is consumed in the pumping which is P_{fr}. So, η_{ov} is expressed as:

$$\eta_{ov} = \eta_{el} + \eta_{th} - P_{fr}/P_{in} \tag{13}$$

2.3. Artificial neural network (ANN) for PVT performance prediction

The ANN is one of the most powerful tools for predicting system performance. There are several factors that influence the selection of ANN, these factors depend on the type and complexity of each problem, and the type and size of the available data. So, it is a complex process to select the appropriate ANN type for a specific data type. Many types of neural networks can be used in the prediction such as CFF (Nait Amar, 2020), FFNN (Esmaeili-Faraj et al., 2021), RBF (Syah et al., 2021), GR (Mahmoodi et al., 2018), and Elman (Vaferi et al., 2015) which are the most widely used of the ANN types.

2.3.1. Radial base function neural network (RBFNN) model

The FFNN is considered as a technique of a supervised learning used in the prediction and the classification problems. Its topology consists of multilayers of which the first layer is the input layer and the last one is the output layer. Also, it has in-between layers called hidden layers. The radial base function neural network (RBFNN) is a feedforward neural network type, where the network is represented as a problem of curve-fitting in high-dimensional data (Abdel daiem et al., 2021). In this type of network, the hidden layers use the RBF as an activation function. The most common use of the RBF is the Gaussian function. The feature vector applies a non-linear transfer function, and this increases the dimension of the feature vector before going to the classification problem. This greatly helps in making the pattern classification problem separable. RBFNN does not apply parameter learning as in the FFNN. It performs linear adjustment of the weights for the radial bases. This characteristic gives the RBFNN the advantage of a very fast converging time without local minima. Moreover, RBF neurons have maximum activation when the center/weights are equal to the inputs. Thus, RBF neural networks are good for novelty detection. It also enjoyed many advantages such as the ease in design, good generalization, a single hidden layer, strong tolerance to input noise, easy to interpret the function of each hidden layer node, and ability to learn using both supervised and unsupervised algorithms (Jia et al., 2014).

The input layer neurons act as the network input pattern, whilst the hidden layers carry out the nonlinear transformations from the input to the hidden variable space. The RBF in the hidden layer can be identified by two factors: the center position and the width parameter. By determining these factors, the input vector can be processed to the hidden spaces. Finally, the output layer handles the hidden neurons outputs and determines the output of the RBFNN (Abdel daiem et al., 2021). The proposed RBFNN flowchart is illustrated in Fig. 6. It has four input parameters (ambient temperature, solar radiation, convection coefficient, and mass flow rate of the cooling fluid). So, the input layer has four neurons. It has five neurons in the output layer (η_{th}, η_{el}, η_{ov}, average temperature of the top surface of the PV panel, and temperature un-uniformity factor).

The output of hidden layer in the RBFNNs depends on the distance between the inputs and the center of the radial base function. It is expressed by using the following relation.

$$Y_l = \sum_{i=1}^{I} \overline{\omega}_{il}\ \varphi_i(\overline{X}, \gamma_i)\quad i = 1, 2, \ldots\ldots I \tag{14}$$

$$\varphi_i = e^{\left(-\frac{\left\|\overrightarrow{X} - \overrightarrow{\delta}_i\right\|}{2\sigma_i^2}\right)} \tag{15}$$

where Y_l is the lth neural network output and $\overline{X}$ is the neural network input vector. $\overline{\omega}_{il}$ is the weights between neuron ith and lth. φ_i is the RBF for the ith neuron, δ_i is the centre of the ith RBF, and I is the total number of neurons in the hidden layer. In the proposed RBFNN, the linear summation function is applied as an activation function for the output neurons while the hidden layer applies the Gaussian function (Dash et al., 2016). σ_i is the width of the ith RBF, and $\left\|\overrightarrow{X} - \overrightarrow{\delta}_i\right\|$ is the norm of the RBF. The output of the RBFNNs can be expressed as;

$$Y_l = \sum_{i=1}^{I} \overline{\omega}_{il} e^{\left(-\frac{\left\|\overrightarrow{X} - \overrightarrow{\delta}_i\right\|}{2\sigma_i^2}\right)}\quad i = 1, 2, \ldots\ldots I \tag{16}$$

The accuracy of the proposed RBFNN depends on the number of hidden neurons. Where, using many neurons may lead to overlearning, and using small ones may lead to failure to reach the exact solution.

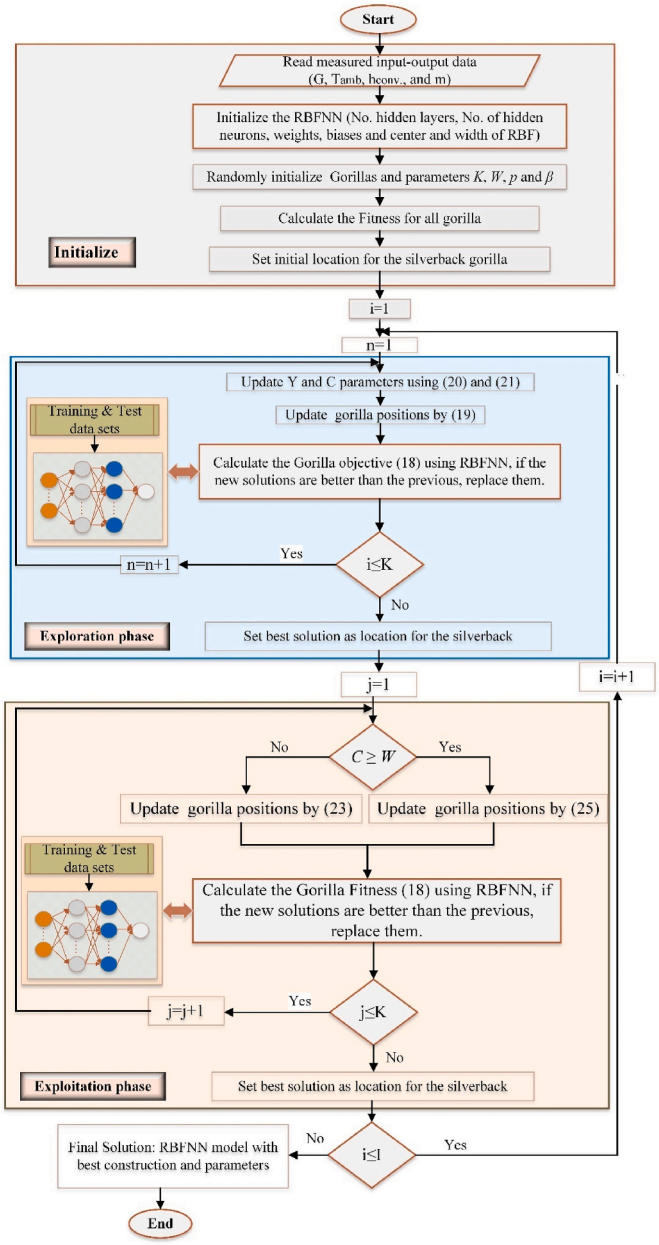


Fig. 7. Flowchart of the proposed GTO-RBFFNN model.

Table 4
Illustrates the RBFNN-GTO parameters.

Parameter	Value
Gorilla Troops dimension	6 × 10
Maximum iteration	300
No. of gorillas	6
Location probability parameter (P)	0.03
β	3
W	0.8

Usually, the trial-and-error method is used to find the number of hidden layers' neurons by starting from a low number of hidden neurons and then increasing them till reach the minimum error for the RBFNN. Recently, new optimization algorithms such as meta-heuristic optimization algorithms have been applied to calculate the optimum construction of the ANNs to satisfy minimum error (Talaat et al., 2022). In the present study, the optimization algorithm is used to find the optimum values of the RBF parameters (centre and width), the connected weights between layers, and the number of hidden neurons. Therefore, the GTO algorithm is implemented to obtain the optimum structure of the proposed RBFNN.

In this paper, minimizing the mean square error (MSE) between the proposed RBFNN output, Z_h, and the actual outputs (Target values), T_h, is used as an objective function for the GTO algorithm. It can be determined using the following equations.

$$\mathrm{MSE} = \frac{1}{\mathrm{N}} \sum_{h=1}^{N} (\mathrm{T_h} - \mathrm{Z_h})^2 \tag{17}$$

$$F_{obj} = \min(\mathrm{MSE}) = \min\left(\frac{1}{\mathrm{N}} \sum_{h=1}^{N} (\mathrm{T_h} - \mathrm{Z_h})^2\right) \tag{18}$$

where F_{obj} is the objective function, N is the number of training patterns,

2.3.2. Gorilla troops optimizer (GTO) algorithm

As a new natural meta-heuristic optimization algorithm, GTO simulates the natural group behavior of the gorilla troops (Abdollahzadeh et al., 2021). It is applied for solving engineering optimization problems due to its advantages. It has numerous advantages such as ease of implementation and layout, needs no information for derivation functions, and performs more than local search algorithms. GTO is built depending on five agents to carry out the exploration and exploitation stages in optimum solution search. The exploration phase includes three agents; migration to a known location, changing the position to other gorillas, and emigration to unexplored locations. These operators enhance the searchability of the GTO and attain more equilibrium between exploration and exploitation. Furthermore, the exploitation includes two agents; follow the silverback and compete for adult females (Abdollahzadeh et al., 2021), (Hatata et al., 2022). In the following sections, the mathematical model of the Gorilla Troops behavior is described.

2.3.2.1. Exploration phase. According to the way of life of the gorilla group, the exploration phase includes three agents; migration to a known location, change position to other gorillas, and migration to unexplored locations. Thus, the gorillas in this algorithm can be considered as possible solutions, and the silverback gorilla represents the optimum location for each optimization iteration (Hatata et al., 2022). For the emigration to an unexplored location is demonstrated as variable. Therefore, three conditions are tested for the implementation of one of the three mechanisms relying on the value of a random variable, R. The emigration to an unexplored location mechanism is selected when the variable R is less than the P value. Changing position to other gorillas is chosen when the R variable is larger than or equal to 0.5. Finally, if R is less than 0.5, emigration to a known location is selected. These agents are expressed as;

$$GL(i+1) = \begin{cases} (B_{max} - B_{min}) \times r_1 + B_{min} & R < P \\ (r_2 - C) \times L_r(i) + Y \times H & R \geq 0.5 \\ L(i) - Y \times (Y \times (L(i) - GL_r(i)) + r_3 \times (L(i) - GL_r(i))), & R < 0.5 \end{cases} \tag{19}$$

where $GL(i)$ represents the possible vector solutions for the gorilla locations for ith iteration and $GL(i+1)$ is the possible solutions for gorilla locations in $(i+1)^{th}$ iteration. B_{min} and B_{max} are the minimum and maximum values of variables. X_r is a gorilla chosen randomly from the group. r_1, r_2, r_3, and R denotes varied random values between 0 and 1. Parameters C, Y, and H can be calculated by,

$$C = (\cos(2 \times r_4) + 1) \times \left(1 - \frac{i}{I}\right) \tag{20}$$

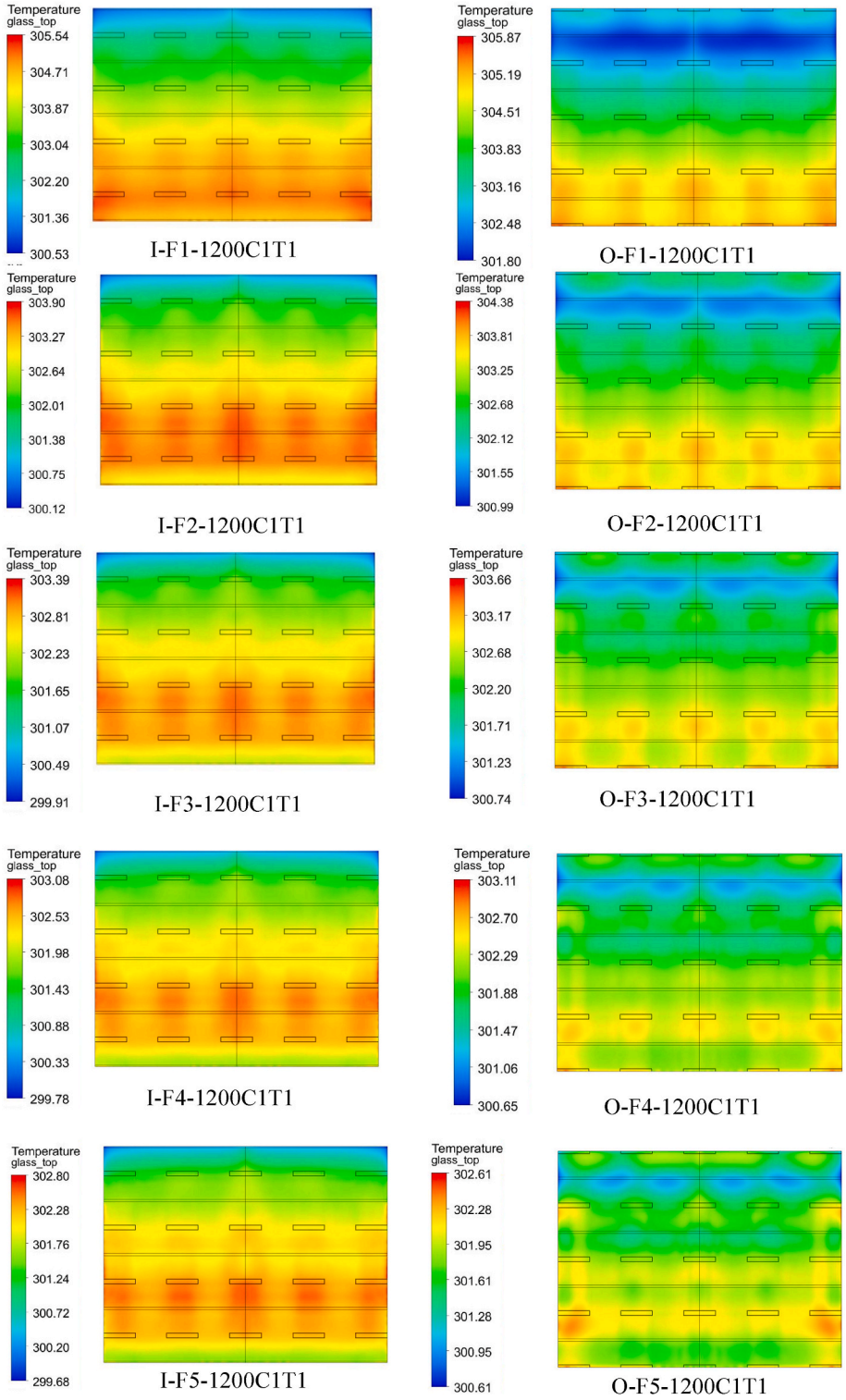


Fig. 8. Variation of the PV temperature in [K], under solar radiation of 1200 W/m^2.

$$Y = \left((\cos(2 \times r_4) + 1) \times \left(1 - \frac{i}{I} \right) \right) \times l \tag{21}$$

$$H = z \times L(i) \tag{22}$$

where I is the maximum iteration number. Z, l and r_4 are randomly selected variables between [-C, C], [-1, 1], and [0, 1] respectively. Equation (19) models the leadership of the silverback. The fitness value of the gorilla location is attained at the exploration phase ends at each iteration, if the gorilla location, *L(i),* is larger than the fitness value, *GL (t),* then the fitness value will be replaced with L (i). So, the finest solution selected in this phase could be indicated as a Silverback.

2.3.2.2. Exploitation phase. Two mechanisms are performed in this phase to obtain the optimal solution; follow the silverback and compete for adult females. The selection of the appropriate mechanism depends on the value of parameter C. The mechanism "Follow the silverback" is

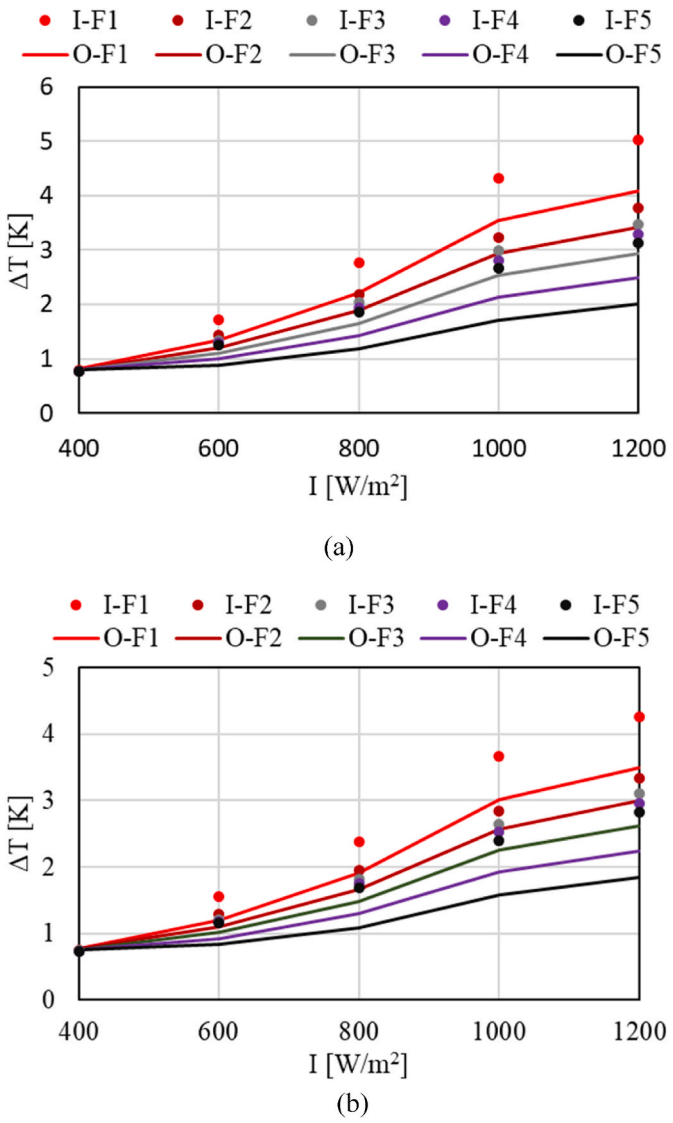


Fig. 9. Temperature un-uniformity for the test cases for a) Convection coefficient 5.84 W/m^2.k, and b) Convection coefficient 20.7 W/m^2.k.

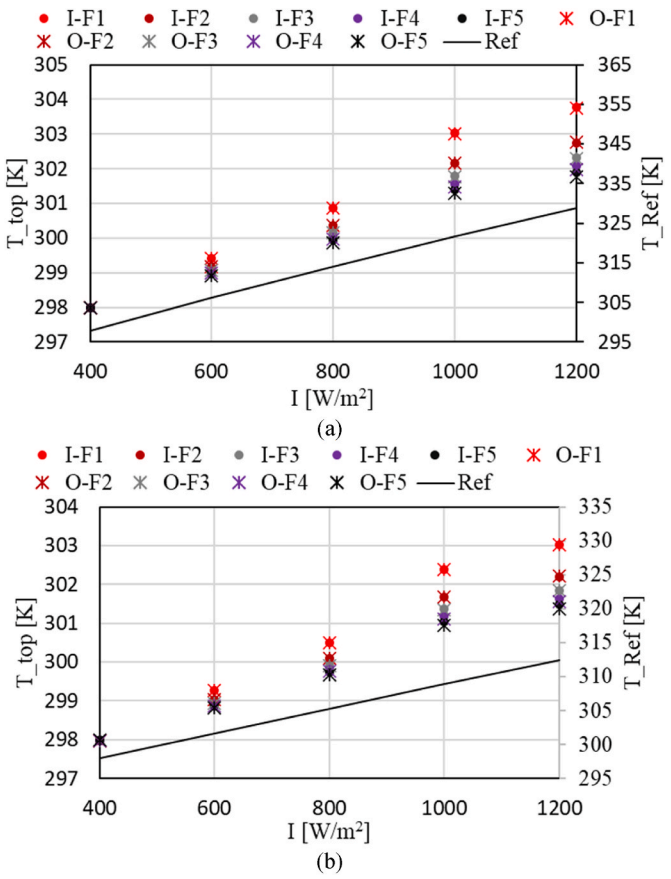


Fig. 10. PV Average temperature for the test cases of a) Convection coefficient 5.84 W/m^2.k, and b) Convection coefficient 20.7 W/m^2.k.

selected when the *C* parameter is larger than or equal to the *Y* parameter. On the other hand, if *C* is less than *Y*, the mechanism "compete for adult females" is chosen.

For recently created gorillas' groups, the silverback is strong and young. So, all gorillas follow the silverback orders. This can be represented by;

$$GL(i+1) = L \times M \times (L(i) - L_{sb}) + L(i) \tag{23}$$

$$M = \left(\left| \frac{1}{K_t} \sum_{j=1}^{K_t} GL_j(t) \right|^{2^Y} \right)^{\frac{1}{2^Y}} \tag{24}$$

where K_t is the total number of gorillas and L_{sb} is the silverback best location. As soon as young gorillas reach adulthood, we find that they begin to compete with other adult males, to expand their influence in selecting adult females. This mechanism is expressed as;

$$GL(i) = L_{sb} - (2 \times r_5 - 1) \times (L_{sb} - L(i)) \times \beta \times E \tag{25}$$

where *E* is the effect of fierceness on the dimensions of solutions. The fitness value of the gorilla location is attained at the exploitation phase end. At each iteration, if the gorilla location, L(i), is larger than the fitness value, *GL(t),* then the fitness value will be replaced with *L* (*i*). So, the finest solution selected in this phase could be indicated as a Silverback.

2.3.3. Hybrid RBFNN-GTO prediction model

PVT power, voltage, current, η_{el}, and η_{el} are affected by many factors such as ambient temperature, the mass flow rate of the cooling fluid, solar irradiance, dust, humidity, and other factors. To model and predict the proposed PVT system, the hybrid RBFNN-GTO is designed to calculate the thermal, electrical, and overall efficiencies. Moreover, the average temperature of the top surface of the PV panel, T_{top}, and the temperature un-uniformity factor, are predicted. The proposed PVT prediction model is based on a numerical dataset.

The prediction model uses the GTO algorithm for training and testing the proposed RBFNN model. It is used to obtain the optimal parameters of the RBFNN by determining the optimum values of the connection weights and biases of each neuron, the number of neurons in the hidden layer, the RBF parameters, and the activation function of each layer. The input data used to train and test the RBFNN-GTO model consists of solar irradiation, ambient temperature, convection coefficient, and mass flow rate. The RBFNN-GTO model is designed, created, and simulated using MATLAB program (R2018). The detailed steps for developing the proposed RBFNN-GTO are as follows.

Step 1. Producing the training and testing data for the RBFNN-GTO model from the parametric database of the numerical model. Then these input-output data sets are fed to the MATLAB program to be processed.

Step 2. Processing the input-output data patterns by normalizing them (irradiation, ambient temperature, conv, and water mass flow) to be within the range [-1, 1]. Normalizing the input-output data sets has the

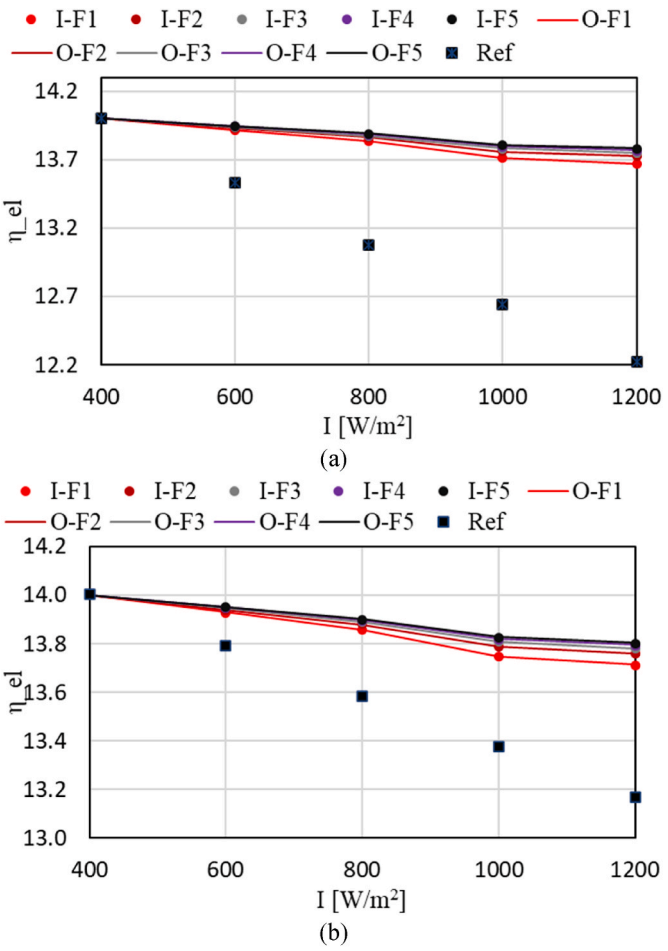


Fig. 11. The values of η_{el} of the PVT model for the test cases of a) Convection coefficient 5.84 W/m^2.k, and b) Convection coefficient 20.7 W/m^2.k.

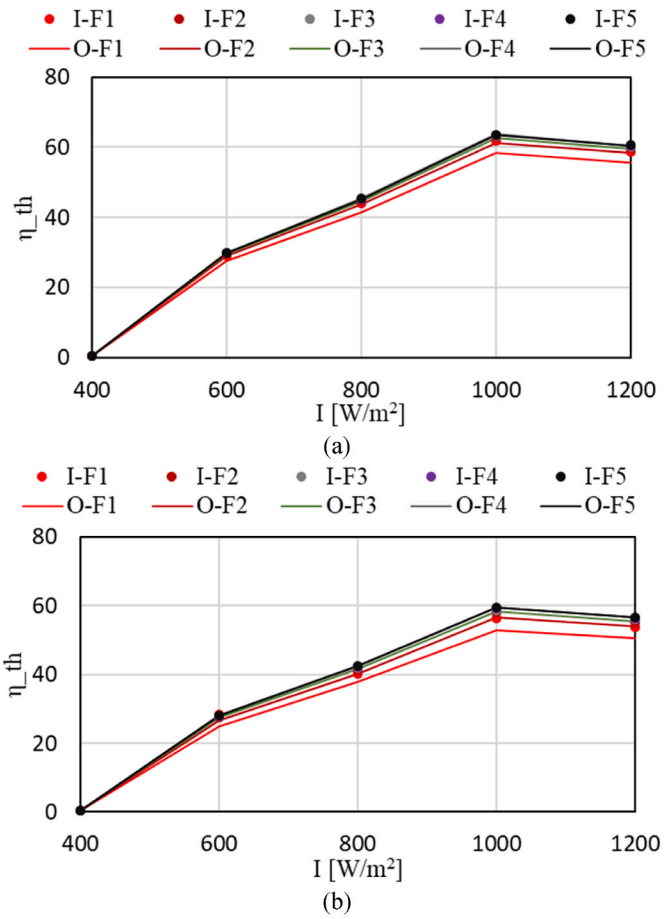


Fig. 12. The values of η_{th} of the PVT model for the test cases of a) Convection coefficient 5.84 W/m^2.k, and b) Convection coefficient 20.7 W/m^2.k.

advantage of improving the proposed RBFNN performance.

Step 3. Initializing the parameter values of the proposed RBFNN randomly, including population size, number of neurons in the hidden layer, the RBF parameters, and the connection weights and biases.

Step 4. Designing the proposed RBFNN with the previous initial parameters.

Step 5. Calculating the fitness function (MSE) using Eq. (18) for each search gorilla in the search data.

Step 6. Applying the GTO algorithm, as shown in Fig. 7, to train the proposed RBFNN to calculate the optimal construction to achieve minimum MSE.

Step 7. Repeat the steps from four to seven until reach the desired value of the MSE or the maximum iteration.

The design details of implementing the proposed GTO algorithm for finding the optimum construction of the RBFNN are illustrated in the flowchart in Fig. 7. As shown in the figure, the GTO improves the RBFNN architecture and its parameters. The MSE function is used in this paper as the gorilla fitness value.

In this research, the training process's initial learning rate is chosen to be 0.0005. Moreover, all the training and testing processes to check the proposed GTO performance are obtained using 40 populations for 300 maximum iterations. The population size, L_i, is initialized randomly and generated to determine the optimal parameters of the proposed RBFNN. Furthermore, the parameters of GTO are illustrated in Table 4. The proposed GTO algorithm has 6 gorillas, and each gorilla is a vector of size 10. So, the Gorilla Troops dimension is 6 × 10 (see Table 5).

3. Results and discussion

The results of the numerical simulation are extracted to compare the performance of the systems with their parameters at different operating conditions. In this section, the temperature distribution of the PV panel is discussed to account for cooling and temperature homogeneity to avoid thermal stresses on the PV module's surface. Furthermore, the η_{th}, η_{el}, and η_{ov} of the system are discussed to show the enhancement achieved by the present cooling technique compared to the uncooled case.

3.1. PV temperature contours

The temperature variation of the PV module surface shows the effectiveness of the cooling process and the homogeneity of the applied cooling technique. Fig. 8. Shows the temperature contours for a sample of the simulation runs performed under solar radiation of 1200 W/m^2, convection coefficient of 5.84 W/m^2.k, and ambient temperature of 298K at five different flow rates. As can be observed from Fig. 6., the minimum temperature of the PV panel in the I arrangement is always smaller than that of the O's arrangement. This refers to that in the I arrangement, the HTF starts the injection at the beginning of the plate. This guarantees that the HTF is not to be preheated before getting in contact with the thermal module. Thus, it achieves smaller temperatures. However, in the O arrangement, the injection of the HTF takes place after passing one stage. During this stage of channels, the HTF is preheated from the fluid beneath it. The maximum temperature is found to be smaller for the I arrangement compared to the O arrangement, for

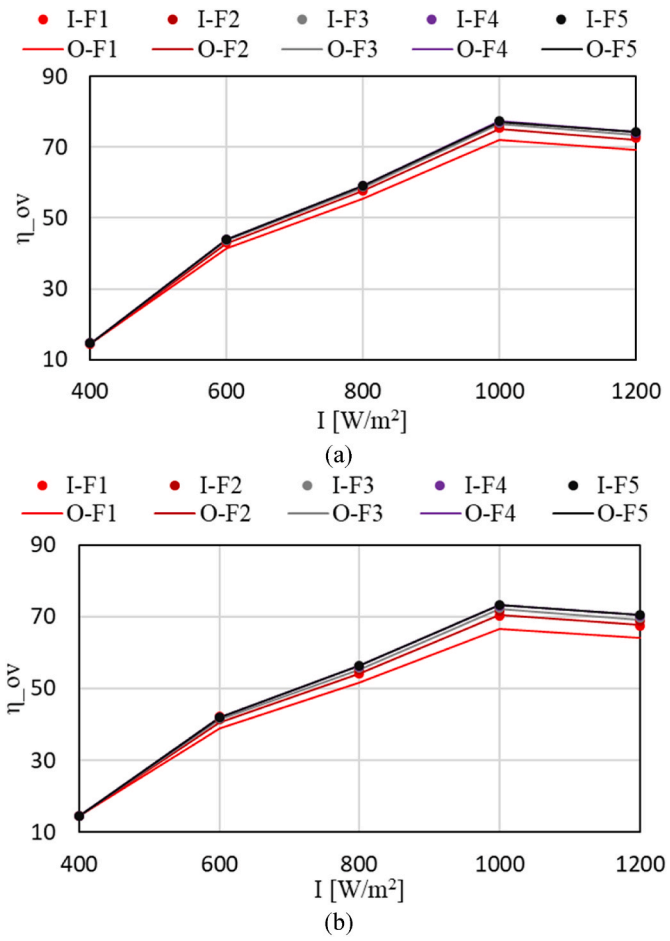


Fig. 13. The values of η_{ov} of the PVT model for the test cases of a) Convection coefficient 5.84 W/m^2.k, and b) Convection coefficient 20.7 W/m^2.k.

most of the cases. This refers to that in the O arrangement, the ability to cool becomes less as the entering fluid temperature increases. So, the maximum temperature is higher as well. In this stage of the data analysis, it can be observed that the ability of cooling for the I arrangement is higher than that of the O arrangement. Furthermore, it can be observed that the cooling efficiency increases with the increase of the flow rate from F1 to F5 due to the increase of the forced convection. Furthermore, it can be observed that the inlet of the cooling HTF is delayed in the O than in the I arrangement. This keeps the temperature as low as possible because the total fluid path decreases in the O arrangement. This results in lower temperature variation in the O arrangement than in the I arrangement.

3.2. PV temperature un-uniformity

The temperature un-uniformity is defined as the difference between the minimum and the maximum temperature of the PV module. This parameter has a crucial effect on the thermal stresses applied to the PV module causing cracks. The temperature difference variation according to the different tested cases is shown in Fig. 9.

It can be observed from Fig. 9 that the temperature un-uniformity in the O configuration is less than in the I configuration, especially for high radiation. This may refer to the delay of the inlet in the O than in the I geometry. This in turn keeps the fluid at a lower temperature after some length of the panel before getting in contact with the PV module back. This causes the fluid path length to be smaller in the O than in the I configuration. As a result, the cooling process takes place with a lower maximum temperature in the O arrangement. Furthermore, it is found that as the flow rate increases from F1 to F5, the temperature un-uniformity decreases. This refers to the increase of the forced convection with the flow rate increase.

3.3. PV average temperature

The average temperature of the PV module for the test cases is illustrated in Fig. 10. It can be observed that the average temperature increases with the irradiance power and decreases with the HTF flow rate. Moreover, the average temperature for both O and I arrangements is found to be identical for the first four flow rates. As the flow rate reaches F5, the O arrangement achieves a little bit smaller temperature than the I arrangement by 0.14 K and 0.12 K, for convection coefficients of 5.84 and 20.7 W/m^2.k, respectively. The proposed cooling achieves a decrease in the PV temperature by up to 27K and 11K for convection coefficients of 5.84 and 20.7 W/m^2.k, respectively.

3.4. Electrical efficiency

The variation of η_{el} of the system for the different test cases is illustrated in Fig. 11. It is known that η_{el} of the PV panels is proportional to its average temperature. So, the O arrangement which achieves lower temperature achieves higher η_{el} as shown. It can be observed from Fig. 11., that the difference in η_{el} between the O and I arrangements is declared only at the highest HTF flow rate F5. As the irradiance power increases, it is found that η_{el} decreases as well due to the PV temperature increase. As the flow rate makes cooling for the PV module, η_{el} is proportional to the flow rate. The effect of the external convection coefficient is proportional to η_{el} as it decreases its temperature, as well.

The proposed cooling techniques achieve an enhancement in η_{el} by up to 12.7% and 4.8% for convection coefficients of 5.84 and 20.7 W/m^2.k, respectively.

3.5. Thermal efficiency

The variation of η_{th} for the test cases is illustrated in Fig. 12. It can be observed that η_{th} is proportional to the flow rate. This refers to the increase of the heat transfer by convection with the flow rate. The external convection is found to inversely affect η_{th} as it helps in cooling the PV panel and losing some heat to the ambient. For the low irradiance power of 400 W/m^2, η_{th} ranges between 0.41% and 0.54% for the test cases. Furthermore, it can be observed that as the irradiance power increases, η_{th} increases except for the highest irradiance power of 1200 W/m^2.

This refers to the increase of thermal energy losses by convection and radiation as the PV panel's temperature increases. These thermal losses are proportional to the panel's temperature. At a specific level of temperature, the collected thermal energy concerning the irradiance energy received starts to decrease. This effect is very common in solar thermal collectors, and it depends on the degree of thermal insulation of the system. It can be observed that η_{th} of the I arrangement is better than that of the O arrangement. Moreover, it is observed that the effect of the flow rate in the O arrangement is higher than that in the I arrangement. The PVT system under consideration achieves η_{th} of up to 63.55%.

3.6. Overall efficiency

The variation of η_{ov} of the PVT system is shown in Fig. 13.

It can be observed that η_{ov} is affected by η_{th} at high irradiances and by η_{el} at low irradiances at which thermal output is very small. It is observed that the I arrangement achieves higher η_{ov} values, especially for low HTF flow rates. The system understudy achieves η_{ov} up to 77.36% under the test conditions of this study.

3.7. Flow patterns

The flow patterns in the thermal module for both I and O

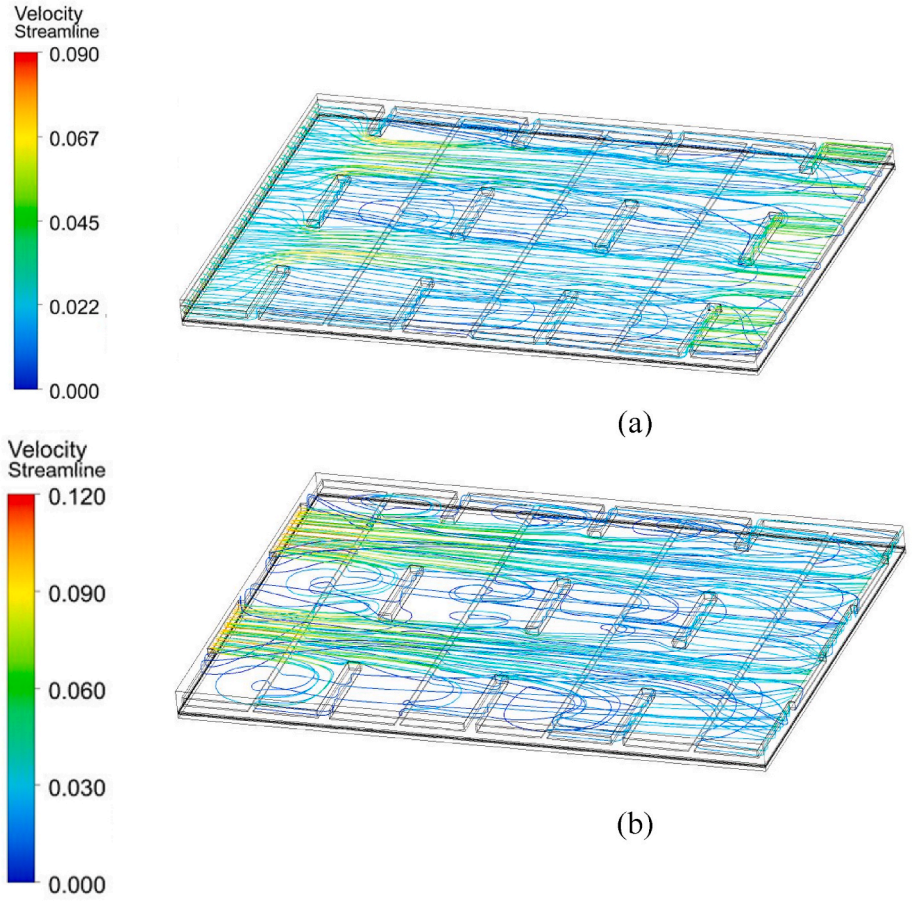


Fig. 14. Velocity streamlines at 1200 W/m^2.k Irradiance power, and F5 flow rate for a) I arrangement, and b) O arrangement.

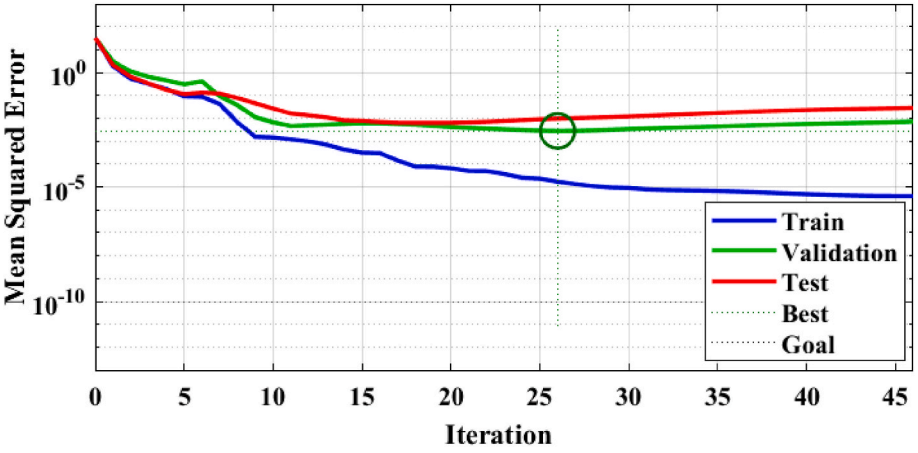


Fig. 15. RBFNN-GTO training performance.

arrangements is shown in Fig. 14. For the I arrangement, the cooling process starts directly contacting the PV back side as it enters the thermal module. The rest of the fluid continues to enter from the next entrance channel passing between the exit channels. For the I arrangement, the streamlines appear to be covering the whole area compared to the O arrangement. This may refer to the small area available for the entering fluid in the O arrangement. This, in turn, restricts the entrance of the fluid from the entrance channels at the locations of the high fluid speeds (between the exit channels) more than the regions behind the exit channels. This causes in-homogeneity for the fluid distribution in the O arrangement compared to the I arrangement. This may be the cause of the higher thermal and η_{ov} achieved by the I arrangement compared to the O arrangement.

3.8. Application of RBFNN-GTO model

In this research, the hybrid RBFNN-GTO model is optimally designed for investigating the performance of cascade channels' cooling for PV modules. The RBFNN-GTO model used the flow rate, convection coefficient, ambient temperature, and solar irradiation as the input parameters to train and test the proposed RBFNN using the GTO method. The input data used to train the RBFNN-GTO model is 93 patterns, which are

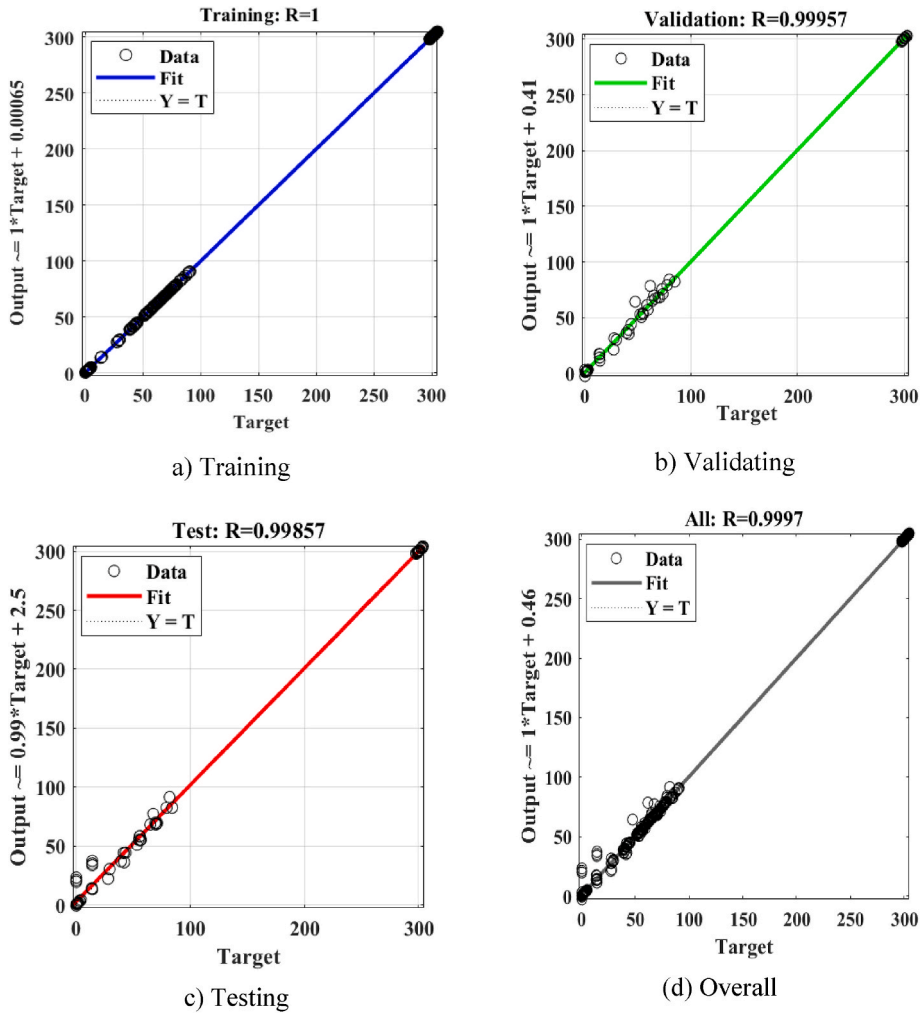


Fig. 16. Correlation between the output values of the RBFNN-GTO and actual output values.

determined from simulating the PVT system under various values of the input parameters as illustrated in Table 3. These dataset patterns used to train the RBFNN-GTO are split into three portions: training dataset with 70% of total measured data, validating dataset with 15% of total measured data, and testing dataset with 15% of total measured data. The RBFNN and GTO are programmed on a Laptop with an Intel (R) (TM) i7-7500U CPU @ 2.70 GHz 2.90 GHz and 16 GB RAM.

The optimum design of the proposed RBFNN-GTO consists of 17 neurons for the hidden layer. This RBFNN-GTO minimizes the MSE to less than $3.7613^{*}\ 10^{-4}$ within 26 iterations. Fig. 15 illustrates the convergence curves of testing, validating, and training the proposed RBFNN-GTO. Also, Fig. 16 shows the regression factor, *R*, for testing, validating, and training the proposed RBFNN-GTO. The regression factor value is equal to 0.9997 which is very close to one. This denotes that, the proposed RBFNN-GTO model can effectively estimate the η_{th}, $\eta_{el,}$ and η_{ov}, the average temperature of the top PV surface, and the temperature un-uniformity factor. Furthermore, the GTO determines the optimum structure of the RBFNN model.

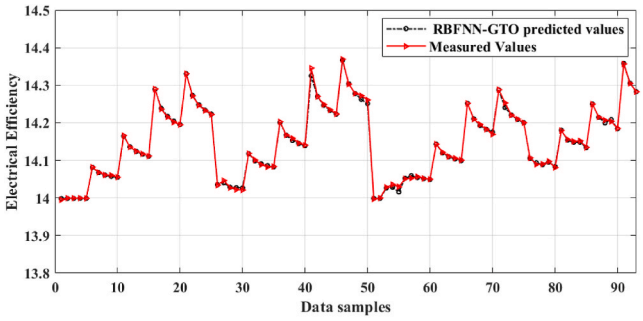
A comparison between the RBFNN-GTO predicted output parameters values and the measured values for the training datasets patterns (93 patterns) is illustrated in Fig. 17.

After obtaining the best structure of the proposed RBFNN-GTO and its parameters, its performance is tested with new datasets which are not used for training it. The objective of this test is to check the ability of the RBFNN to generalize and extrapolate. The proposed RBFNN-GTO is tested using different cases of 15 patterns. Fig. 18 illustrates the comparison between the predicted values of η_{th}, η_{el}, η_{ove}, the average temperature of the PV surface, and the temperature un-uniformity factor from the RBFNN-GTO model and the actual output values, which showed its ability to correctly predict the correct values.

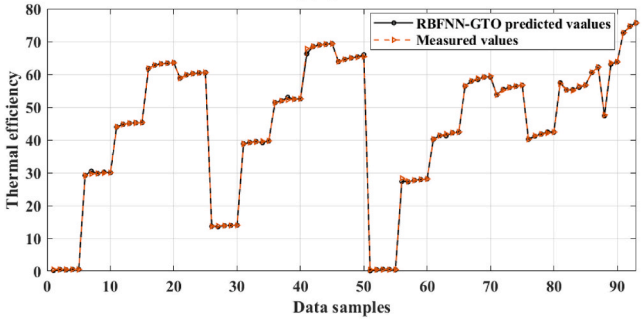
The performance of the RBFNN-GTO model in predicting the performance of the PVT is evaluated using the MSE and R. Table 5 illustrates the MSE and R values for training and testing the proposed RBFNN-GTO model. These values are compared with different neural networks under the same conditions. These networks are (FFNN), RBFNN trained with backpropagation algorithm (BP), and FFNN trained with ant colony optimization (ACO) method (Beltramo et al., 2016). It can be noticed that the RBFNN-GTO achieves better performance. This is because of the high ability of the GTO to obtain the optimum network parameters and structure (see Table 5).

4. Economic study

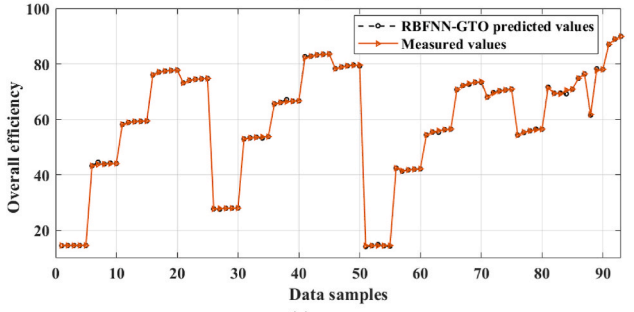
An economic study is performed to show the economic impacts of the modified PV cooling in this article. The flow rate of the cooling is F5. The calculations of the economic study used the equations in (Yanhua et al., 2019). The study is shown in Table 6. The average hours of the day are considered as 6 h and the average irradiance power is $600 W/m^2$. As can be observed from the table, the PVT systems provide lower price for the electric energy generated over the year by 9.4%. considering the extra cost paid for the thermal module. The cost of the thermal energy produced by the PVT systems is 0.0096 $/kW.hr for both the I and O



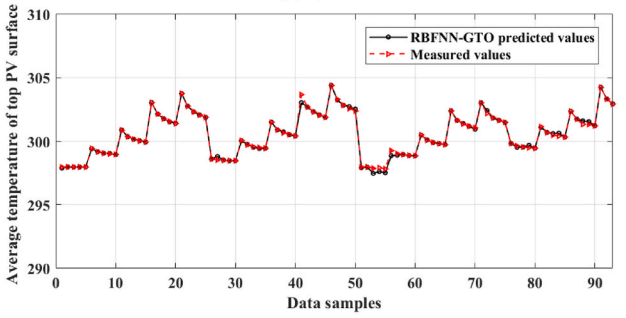
(a) η_{el}



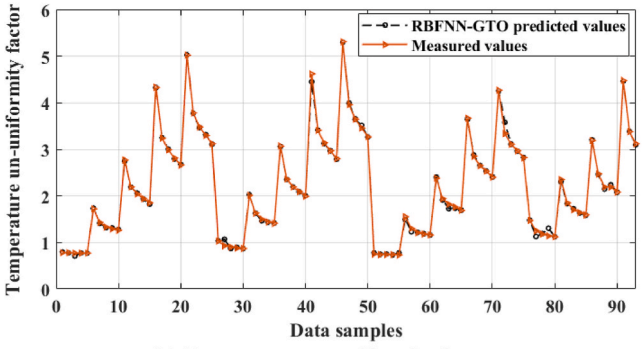
(b) η_{th}



(c) η_{ov}

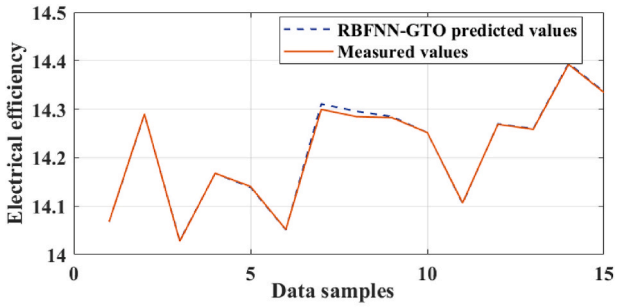


(d) Average temperature of the top PV surface

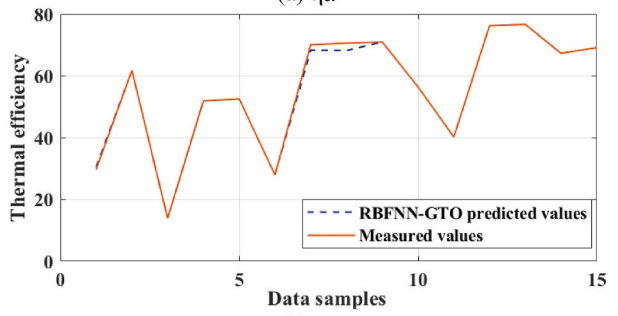


(e) Temperature un-uniformity factor

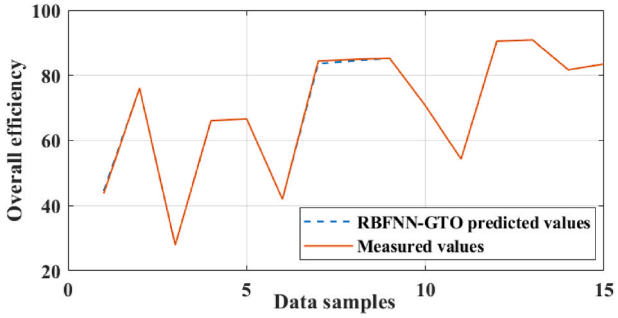
Fig. 17. Comparison between the RBFNN-GTO predicted values and the measured values for the training datasets patterns.



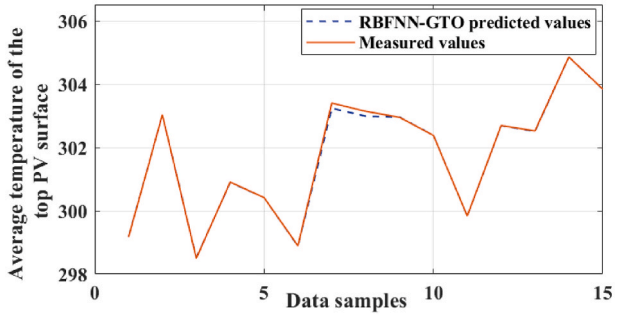
(a) η_{el}



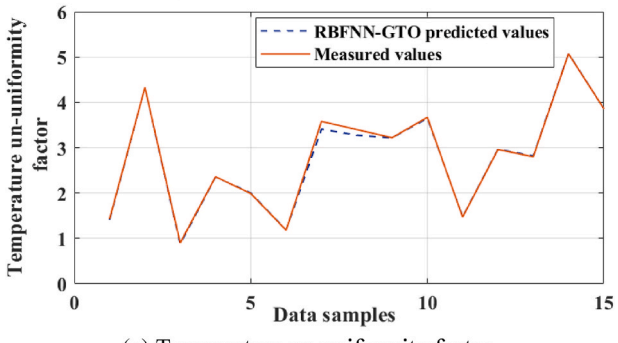
(b) η_{th}



(c) η_{ov}



(d) Average temperature of the top PV surface



(e) Temperature un-uniformity factor

Fig. 18. Comparison between the RBFNN-GTO and measured outputs for the testing data.

Table 5
Different neural network performance evaluation.

Model	Training		Testing	
	R	MSE	R	MSE
RBFNN-BP	0.9987	2. 6412* 10^{-3}	0.9981	6.9812* 10^{-3}
FFNN-BP	0.99901	9.7613* 10^{-4}	0.9985	8.6733* 10^{-3}
FFNN-ACO	0.9992	7.3206* 10^{-4}	0.9989	9.8667* 10^{-4}
RBFNN-GTO	**0.9997**	**3.7613* 10^{-4}**	**0.9991**	**5.4313* 10^{-4}**

Table 6
Cost analysis for the PV system without cooling, and the PVT systems with I and O orientation of the cooling channels.

		PV	PVT with I	PVT with O
Parameter	Unit	Value		
Principal cost (P)	$	50.00	75.00	75.00
Salvage value (S) (10% of principle value)	$	5.00	7.50	7.50
Life of the solar still (n)	year	15.0	15.0	15.0
Interest rate (i)	%	12.0	12.0	12.0
Capital recovery factor (CRF)	–	0.147	0.147	0.147
Sink fund factor (SFF)	–	0.027	0.027	0.027
First annual cost (FAC=CRF × P)	$	7.341	11.012	11.012
Annual salvage value (ASC=SFF × S)	$	0.134	0.201	0.201
Annual maintenance cost (AMC = 0.05 × FAC)	$	0.367	0.551	0.551
Annual cost AC (AC=FAC + AMC-ASV)	$	7.574	11.361	11.361
Electric efficiency		12.600	13.940	13.940
Thermal efficiency		0.000	30.030	29.970
Average daily electric energy yield (Pel*6hr*η_el/A)	kW.hr/m2	0.454	0.502	0.502
Average daily thermal energy yield (Pel*6hr*η_th/A)	kW.hr/m2	0.000	1.081	1.079
Annual yield of electric energy	kW.hr/m2	165.564	183.172	183.172
Annual yield of thermal energy	kW.hr/m2	0.000	394.594	393.806
Cost per electric KW.hr	$/kW.hr	0.0457	0.0414	0.0414
Cost per thermal KW.hr	$/kW.hr	–	0.0096	0.0096
Cost per kW.hr	$/kW.hr	0.0457	0.0197	0.0197

orientation of the cooling channels. This thermal energy is a byproduct of the electric energy production of the PV system. This will reduce the thermal energy provided for heating water using fossil fuel. So, the CO_2 emissions globally will be reduced with these systems.

5. Conclusion

A parametric study was conducted to investigate the performance of a cascade channel cooling PVT system numerically. The investigation test cases included insolation range from 400 to 1200 W/m^2, HTF flow rate from 0.00024 to 0.0012 LPM, external convection coefficients of 5.48 and 20.7 W/m^2.K, and ambient temperatures of 298 and 308 K. Two flow arrangements for the rectangular channels were introduced. One started with the inlet channels, and the other started with the outlet channel. The conclusions from this work are as follows.

- The geometry of the PVT system with the outlet channels first achieved a smaller average temperature of the PV panel by up to 0.14K compared to the inlet first geometry.
- The geometry of the PVT system with the outlet channels first achieved temperature un-uniformity through the PV panels smaller than the inlet first arrangement by up to 1.1K.
- The PVT system under study attained a decrease in the PV temperature by 27K for the low convection and 11K for the high convection cases.
- The enhancement achieved by the proposed PVT system for electrical efficiency reached 12.7% and 4.8% for the low and high wind convection cases, respectively.
- The Thermal efficiency of the proposed system reached up to 63.6% and 59.5% for the low and high wind convection cases, respectively. The I arrangement achieved higher thermal efficiency values than the O arrangement.
- The overall efficiencies achieved by the proposed PVT system were 77.46% and 73.33% for the low and high wind convection cases, respectively. The maximum overall efficiency was achieved under the inlet first arrangement. This referred to the more homogeneous flow profile in the I arrangement causing higher heat transfer rates.
- The PVT systems provided a reduction in the price of the produced electrical energy by 9.4% achieving a price for the produced thermal energy of 0.0096 $/kW.hr calculated from the economic analysis.
- The cooling water coming out from the PV cooling cannot be used as a hot water supply for residential applications because of its low grade. However, some of the solution presented to use this low-grade heat is as a source in heat pumps or as a preheating source for solar heaters that reduces the energy consumed in the main heating process.
- The proposed RBFNN-GTO model presented accurate results and good performance for predicting the electrical thermal and overall efficiencies, the average temperature of the PV surface, and the temperature un-uniformity factor of the PVT.
- Implementing the proposed GTO algorithm led to attaining the optimum structure and parameters of the RBFNN. Thus, it minimized the MSE to 3.7613* 10^{-4}.
- The overall (testing, validating, and training) correlation coefficient was very close to one (0.9997).
- The accuracy of the proposed RBFNN-GTO model was investigated compared with various neural networks.

CRediT authorship contribution statement

Mohamed A. Essa: Methodology, Writing – original draft, Software, Validation, Data curation. **AbedAlrzaq Alshqirate:** Formal analysis, Methodology, Writing – review & editing. **A.Y. Hatata:** Methodology, Writing – original draft, ANN model software, Validation, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

Acknowledgment

The author would like to thank the Deanship of Scientific Research at Shaqra University for supporting this work.

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